

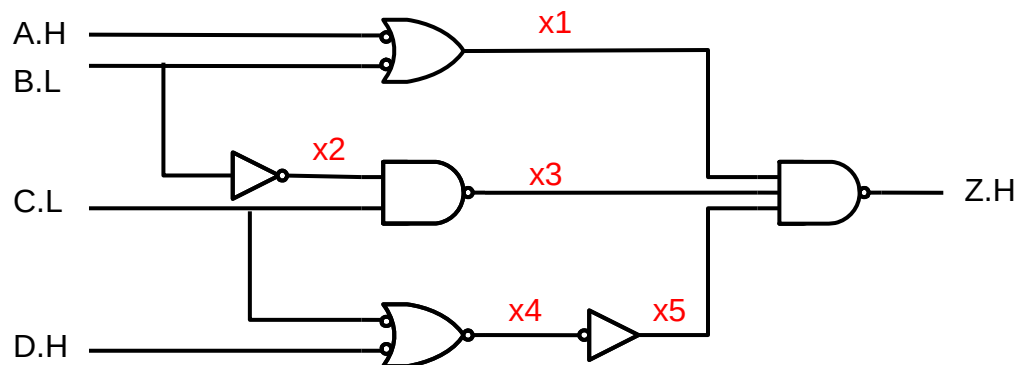
## EE2020/EE2026 (Part 1)

### Tutorial 3 - Questions

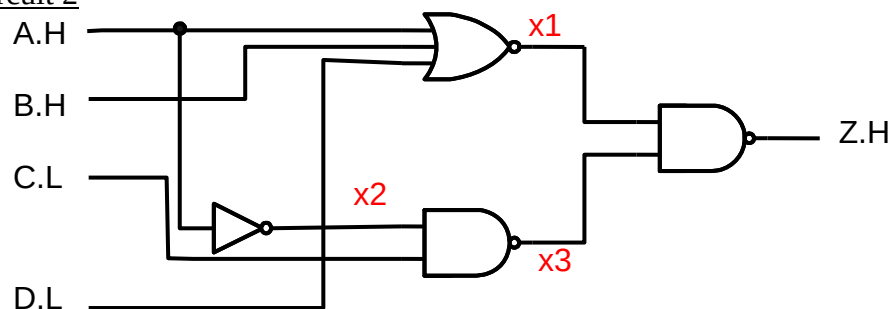
#### Logic gates

1. Use algebraic manipulation to find the MSOP for  $f = x_1x_3 + x_1\bar{x}_2 + \bar{x}_1x_2x_3 + \bar{x}_1\bar{x}_2\bar{x}_3$ .  
Write the Verilog code that describes the above function as a module (use dataflow description style).
2. For circuits 1 and 2 below:
  - (a) Fill in all the intermediate signal names using the positive logic convention
  - (b) Find the logic expression for Z for the two cases
  - (c) Write the Verilog code that describes the above function as a module (use structural description style). In the description, assume that only the Verilog primitives are available (no need to describe): not(out,in), nand(out,in1,in2,...), nor(out,in1,in2,...).

Circuit 1



Circuit 2



3. Design a logic circuit implementing the following Boolean function, using only NOR gates:

$$X = A \oplus B \oplus C$$

where  $\oplus$  represents the 2-operand XOR operation. Use alternate gate representation if necessary for clear circuit diagrams.

Write the Verilog code that describes the above function as a module (use dataflow description style).

4. Design a circuit to realize  $Z = \overline{A}B + \overline{B}\overline{C}D + \overline{B}\overline{D}$  in the positive logic convention.  $A$  and  $B$  are active low signals while  $C$ ,  $D$  and  $Z$  are active high. Use a minimum number of gates, and draw clear circuit diagrams with alternate gate representations as required.

Write the Verilog code that describes the above function as a module (use dataflow description style).